



Chapter 9

Measurement of Nation Income

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Measurement of Nation Income

(5 hours)

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Gross domestic product (Components, real and nominal gross domestic product)

GDP is a broad monetary measure of a nation's overall economic activity, valuing all the final goods and services produced in a particular period of time, typically annually or quarterly, within the country's boundaries. Within each country, GDP is usually measured by a national governmental agency.

GDP is considered to be one of the principal indicators in economics, allowing analysts to build a better picture of a nation's financial situation. A significant change in GDP, whether negative or positive, usually reflects in the stock market.

When the economy is doing well, wages increase and a lower unemployment rate is indicated as businesses demand more labor to meet the growing economic needs. Rising GDP signifies that incomes within the country are increasing respectively as well as consumers purchasing power, and vice-versa.

Gross domestic product (Components, real and nominal gross domestic product)

GDP has a large impact on nearly everyone within that economic environment. It can affect everything from personal finances to investments to job growth.

Understanding the GDP of a given nation may be helpful for investors when making any decisions in a particular region. It is also vital when comparing a country's growth rates to find the best international opportunities. For example, an investor may choose to buy shares in companies that are located in a rapidly growing economy in the hope of higher returns.

GDP is commonly expressed by the formula:

$$GDP = C + I + G + (X - M)$$

Gross domestic product (Components, real and nominal gross domestic product)

I. Consumption (C)

Consumption means spending by households on goods and services.

Examples:

- Food
- Clothes
- Rent
- Transport
- Medical services
- Education fees

It is usually the largest component of GDP in many countries.

2. Investment (I)

Investment means spending on capital goods that help produce other goods and services in the future.

Examples:

- Purchase of machinery
- Construction of factories
- Building houses
- Tools and equipment
- Business inventory

In GDP, Investment does not mean buying shares or bonds. It means spending on physical productive assets.

Gross domestic product (Components, real and nominal gross domestic product)

3. Government Expenditure (G)

Government expenditure means spending by the government on public goods and services.

Examples:

- Roads and bridges
- Salaries of teachers and police
- Hospitals
- Defense services
- Public infrastructure

4. Net Exports (X-M)

Net exports are the difference between exports and imports.

Exports (X) = goods and services produced domestically and sold to other countries

Imports (M) = goods and services bought from other countries

So,

Net Export = $X - M$

If exports > imports, net exports are positive

If imports > exports, net exports are negative

Gross domestic product (Components, real and nominal gross domestic product)

Nominal GDP

Nominal GDP is the GDP measured at current market prices of the same year.

It includes changes caused by:

- change in output
- change in price level

So, if prices rise due to inflation, nominal GDP may rise even if production does not increase much.

Suppose a country produces 100 bags of cement at Rs. 500 each.

$$\text{Nominal GDP} = 100 \times 500 = \text{Rs. } 50,000$$

If next year price rises to Rs. 600 and output remains 100:

$$\text{Nominal GDP} = 100 \times 600 = \text{Rs. } 60,000$$

Here GDP increased, but only because of **price increase**, not because of higher production.

Gross domestic product (Components, real and nominal gross domestic product)

Real GDP is the GDP measured at constant prices of a base year. It shows the actual change in production, excluding the effect of inflation. So, real GDP is a better measure of economic growth.

Using the same example, if the base year price is Rs. 500:

Current output = 100 bags

Base year price = Rs. 500

$$\text{Real GDP} = 100 \times 500 = \text{Rs. } 50,000$$

Even if current market price is Rs. 600, real GDP stays based on base-year price.

Gross domestic product (Components, real and nominal gross domestic product)

Nominal GDP	Real GDP
Nominal GDP is the total value of goods and services produced in a country measured using current market prices.	Real GDP is the total value of goods and services produced measured using constant prices of a base year.
Uses current year prices of goods and services.	Uses base year prices, so prices remain constant.
Includes the effect of inflation or deflation. If prices rise, nominal GDP also increases even if production does not change.	Removes the effect of inflation, so it reflects only changes in production quantity.
Shows the total economic value in current money terms.	Shows the actual growth in production of goods and services.
Less accurate for measuring real economic growth because price changes distort the value.	More accurate for measuring economic growth because it isolates real output changes.

Gross domestic product (Components, real and nominal gross domestic product)

No adjustment is made for price changes.	Prices are adjusted using a GDP deflator or base-year prices.
Difficult to compare economic growth over different years because price changes affect values.	Allows meaningful comparison of economic output across years.
Highly sensitive to inflation or deflation.	Not affected by inflation because price changes are removed.
May show increase even if production stays the same but prices increase.	Shows increase only when actual production increases.
Used to measure the current size of the economy in monetary terms.	Used to analyze real economic performance and growth rate.

Gross domestic product (Components, real and nominal gross domestic product)

Actual Dollars (Nominal/Current Dollars)

The actual amount of money physically spent, earned, or received at the time of a transaction.

Impact of Inflation: They do not account for changes in the cost of living over time.

Usage: Used for straightforward cash flow tracking, such as knowing exactly how many bills you need to pay a current bill or salary.

Example: If your salary is \$100,000 this year and stays \$100,000 next year, your "actual dollars" remain the same even if prices for goods rise.

Gross domestic product (Components, real and nominal gross domestic product)

Constant Dollars (Real Dollars)

Dollar values adjusted to a "base year" to remove the effects of inflation.

Impact of Inflation: They reflect purchasing power, allowing you to compare the value of money across different years as if prices hadn't changed.

Usage: Essential for analyzing economic trends like Real GDP or historical income growth to see if you are actually getting "richer" or just keeping up with rising prices.

Calculation: They are calculated by dividing the actual (nominal) dollar amount by a price index, such as the Consumer Price Index (CPI).

Unemployment (Measurement, job search, minimum wage law)

Unemployment occurs when people who are willing and able to work cannot find a job at the prevailing wage rate.

Measurement of Unemployment

Unemployment is measured using the unemployment rate.

Unemployment Rate Formula

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed People}}{\text{Total Labor Force}} \times 100$$

Unemployment (Measurement, job search, minimum wage law)

Labor Force

Labor force includes:

- Employed people (those who have jobs)
- Unemployed people (those actively looking for jobs)

It does not include:

- students not seeking work
- retired people
- people not willing to work

Unemployment (Measurement, job search, minimum wage law)

- Employed people = 90 million
- Unemployed people = 10 million

Labor Force:

$$90 + 10 = 100 \text{ million}$$

Unemployment rate:

$$\frac{10}{100} \times 100 = 10\%$$

Unemployment (Measurement, job search, minimum wage law)

Job search refers to the process where workers spend time looking for suitable employment.

Even in a healthy economy, unemployment exists because workers need time to find the right job.

Reasons for Job Search

Workers may take time to search because:

- they want better salary
- they want better working conditions
- they want jobs matching their skills
- they need time to collect job information

A **minimum wage law** is a government regulation that sets the lowest legal wage employers can pay workers.

It is designed to protect workers from very low wages.

Inflation : Causes, effects and measurement of inflation

Inflation is the sustained, broad-based increase in the general price level of goods and services in an economy over time. It is not just a single price hike but an ongoing trend where the purchasing power of money declines- meaning each unit of currency buys fewer goods and services than before

Core Causes

Economists typically categorize the causes of inflation into three main types:

- **Demand-Pull Inflation:** Occurs when the demand for goods and services exceeds the economy's ability to produce them (often described as "too much money chasing too few goods").
- **Cost-Push Inflation:** Happens when the cost of production (e.g., raw materials like oil or labor wages) rises, forcing businesses to increase prices to maintain profit margins.
- **Built-In Inflation:** A "wage-price spiral" where workers demand higher wages to keep up with rising costs, which then causes firms to raise prices further to cover those higher labor costs.

Causes, effects and measurement of inflation

Measuring Inflation: Inflation is tracked using various indices that monitor price changes for a "basket" of goods and services over time:

- **Consumer Price Index (CPI):** Measures the average change in prices paid by urban consumers for a market basket of consumer goods and services.
- **Producer Price Index (PPI):** Tracks the average change in selling prices received by domestic producers for their output.
- **Core Inflation:** A version of the CPI that excludes more volatile items like food and energy to reveal underlying long-term trends.
- **GDP Deflator:** A broad measure that reflects price changes for everything produced in an economy.

Causes, effects and measurement of inflation

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Causes, effects and measurement of inflation

Inflation Intensity & Degrees

- **Creeping Inflation (1–3%):** Prices rise slowly at about 1–3% per year, which is considered normal and healthy for economic growth. Purchasing power decreases slightly but remains stable.
- **Walking Inflation (3–10%):** Prices increase moderately at around 3–10% per year, making the cost of living noticeably higher. It can signal that the economy is overheating.
- **Galloping Inflation (10–50%):** Prices rise rapidly between 10–50% per year, significantly reducing purchasing power. Savings and fixed incomes lose value quickly.
- **Hyperinflation (more than 50% per month):** Prices increase extremely fast, often exceeding 50% per month, causing money to lose value dramatically. It can lead to economic collapse and loss of trust in the currency.

Measuring Inflation

$$CPI = \frac{\text{Cost of basket in current year}}{\text{Cost of basket in base year}} \times 100$$

$$Inflation\ Rate = \frac{CPI_{current} - CPI_{previous}}{CPI_{previous}} \times 100$$

$$GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$$

Measuring Inflation

Base Year (2023):

Basket cost = Rs. 10,000

CPI = 100

Current Year (2024):

Basket cost = Rs. 11,000

$$CPI = \frac{11000}{10000} \times 100 = 110$$

$$Inflation = \frac{110 - 100}{100} \times 100 = 10\%$$

Equivalence Calculation Under Inflation

- **Future Value (Actual Dollars):** To find what a current amount will be worth in the future given a specific inflation rate.

$$FV = PV \times (1 + f)^n$$

- *Where:* PV = Present Value, f = average annual inflation rate (as a decimal), and n = number of years.

- **Present Value (Constant Dollars):** To find the value of a future sum in today's terms (purchasing power).

$$PV = \frac{FV}{(1 + f)^n}$$

- *Example:* If you receive **\$100,000** in **3 years** with a **3%** inflation rate, its present value is approximately **\$91,514**.

Equivalence Calculation Under Inflation

- **Real Interest Rate (Fisher Equation):** To find the "true" earning power of an investment after removing inflation.

$$i_{real} \approx i_{nominal} - f$$

- *Exact Formula:* $1 + i_{nominal} = (1 + i_{real})(1 + f)$

Equivalence Calculation Under Inflation

Relationship:

$$(1 + i) = (1 + r)(1 + f)$$

Where:

- i = nominal interest rate
- r = real interest rate
- f = inflation rate

Questions

Find rate of inflation per year when the price of a product has increased from Rs. 5,00,000 to Rs. 6,30,000 over a period of 3 years.

Solution:

We know,

$$F = P(1 + r)^N$$

or,

$$6,30,000 = 5,00,000(1 + r)^3$$

Solving,

$$r = 0.08 = 8\% \text{ per year}$$

Questions

Calculate the average (compound) inflation rate when CPI has increased from 176 to 216 over the past 3 years.

Given: Initial CPI (P) = 176

Final CPI (F) = 216

Number of years (N) = 3

We know the compound inflation formula:

- Initial CPI (P) = 176
- Final CPI (F) = 216
- Number of years (N) = 3

We know the compound inflation formula:

$$F = P(1 + f)^N \quad f = \left(\frac{216}{176}\right)^{\frac{1}{3}} - 1 = 0.071 = 7.1\% \text{ per year}$$

Where:

- f = annual inflation rate

Questions

Convert an actual cash flow of Rs. 20,000 per year for 5 years into constant (real) cash flow when inflation rate is 5%.

Also calculate the loss in purchasing power each year.

$$\text{Real Cash Flow} = \frac{\text{Actual Cash Flow}}{(1 + f)^n}$$

Year (EOY)	Actual Cash Flow	Conversion Factor	Constant Cash Flow	Loss in Purchasing Power
0	20,000	$(1.05)^0$	20,000	0%
1	20,000	$(1.05)^{-1}$	19,048	4.76%
2	20,000	$(1.05)^{-2}$	18,141	9.30%
3	20,000	$(1.05)^{-3}$	17,277	13.62%
4	20,000	$(1.05)^{-4}$	16,454	17.73%

Even though you receive Rs. 20,000 every year, its real value decreases because of inflation.

After 4 years, Rs. 20,000 feels like only Rs. 16,454 in today's money.

Questions

Choose the best project from the following alternatives.

Project	Machine X	Machine Y
First cost	15,00,000	20,00,000
Life	7 years	7 years
Salvage Value	2,00,000	3,00,000
Annual operating and maintenance cost	3,00,000	2,50,000

Assume an average inflation of 5% for the next five years and interest rate is 15% per year.

Questions

Effective rate of interest (i_{eff})

$$\begin{aligned}i_{eff} &= (1 + f)(1 + i_{free}) - 1 \\&= (1 + 0.05)(1 + 0.15) - 1 \\&= 0.2075 \text{ or } 20.75\%/year\end{aligned}$$

PW of Project X:

$$\begin{aligned}&= -15,00,000 - 3,00,000(P/A, 20.75\%, 7) + 2,00,000(P/F, 20.75\%, 7) \\&= -25,06,166\end{aligned}$$

Questions

PW of Project Y:

$$\begin{aligned} &= -20,00,000 - 2,50,000(P/A, 20.75\%, 7) + 3,00,000(P/F, 20.75\%, 7) \\ &= -28,02,849 \end{aligned}$$

Since minimum cost is for Project X, **Project X is the best choice.**

Questions

You borrowed Rs. 1,20,000 and agreed to pay Rs. 6,000 per month for 2 years (24 months). If average inflation is 0.75% per month, find the inflation-free (real) interest rate.

- $P = 1,20,000$
- $A = 6,000$ per month
- $N = 24$ months

We use:

$$P = A(P/A, i, N)$$

$$P = A \left[\frac{(1+i)^N - 1}{i(1+i)^N} \right]$$

Substitute:

$$1,20,000 = 6,000 \left[\frac{(1+i)^{24} - 1}{i(1+i)^{24}} \right]$$

Questions

Substitute:

$$1,20,000 = 6,000 \left[\frac{(1+i)^{24} - 1}{i(1+i)^{24}} \right]$$

By solving,

$$i_{nominal} = 0.0151 = 1.51\% \text{ per month}$$

Convert Nominal Rate to Real (Inflation-Free) Rate

We use the standard relation:

Questions

- $i = 0.0151$
- $f = 0.0075$ (0.75%)
- r = real interest rate

Substitute:

$$1.0151 = (1 + r)(1.0075)$$

$$1 + r = \frac{1.0151}{1.0075}$$

$$1 + r = 1.007543$$

$$r = 0.007543$$

$$r = 0.7543\% \text{ per month}$$

Questions

Compute the equivalent present worth (PW) using the inflation method.

Given:

- Cash flows are in Actual Dollars
- Inflation rate $f=5\%$
- Real interest rate $i=10\%$

EOY	Cash Outflow	Cash Inflow	Net Cash Flow
0	-10,00,000	—	-10,00,000
1	1,00,000	5,00,000	4,00,000
2	2,00,000	5,60,000	3,60,000
3	3,00,000	6,20,000	3,20,000
4	4,00,000	6,80,000	2,80,000
5	5,00,000	7,40,000	2,40,000

Questions

- Effective Interest Rate:

$$i_{eff} = (1 + f)(1 + i) - 1 = (1.05)(1.10) - 1 = 0.155 = 15.5\%$$

- Present Worth (PW):

$$PW = -10,00,000 + \sum \frac{\text{Net Cash Flow}_t}{(1.155)^t} = 97,963$$

Questions

Calculate IRR if:

- MARR = 12%
- Inflation rate (f) = 8%
- Cash flows are given in Constant Dollars

Year	0	1	2	3	4
Constant Dollar	-6,000	1,500	2,000	2,500	3,000

Year	Constant	Conversion Factor	Actual Dollar
0	-6,000	—	-6,000
1	1,500	$(1.08)^1$	1,620
2	2,000	$(1.08)^2$	2,332.8
3	2,500	$(1.08)^3$	3,149.28
4	3,000	$(1.08)^4$	4,081.47

Questions

- Convert constant dollars to actual dollars using $(1.08)^t$, then set

$$PW = -6000 + \sum \frac{\text{Actual Cash Flow}}{(1 + i)^t} = 0$$

- By trial:

At 25% → $PW = +54$

At 26% → $PW = -51$

$$IRR = 25.52\% > MARR(12\%)$$

Inflation Controlling Measures

To control inflation, governments and central banks use a combination of monetary, fiscal, and supply-side measures. The primary goal is typically to reduce aggregate demand or increase the economy's productive capacity.

Monetary Measures

- Central banks primarily use monetary policy to manage the money supply and interest rates.
- Interest Rate Hikes: Raising policy rates (e.g., the repo rate or federal funds rate) makes borrowing more expensive, which discourages consumer spending and business investment.
- Increasing Reserve Requirements: By raising the Cash Reserve Ratio (CRR) or Statutory Liquidity Ratio (SLR), central banks force commercial banks to hold more cash, reducing the amount available for lending.
- Open Market Operations (OMO): The central bank sells government securities to absorb excess liquidity from the banking system.
- Quantitative Tightening: Stopping or reversing quantitative easing (buying bonds) reduces the money supply in circulation.

Inflation Controlling Measures

Fiscal Measures

- Governments use fiscal policy to influence the economy through taxation and spending.
- Reducing Government Expenditure: Cutting public spending on infrastructure, defense, or subsidies directly reduces the total demand in the economy.
- Increasing Taxation: Raising direct taxes, such as personal income or corporate taxes, reduces disposable income for households and profits for firms, leading to lower consumption and investment.
- Deficit Reduction: Moving toward a budget surplus helps lower long-term interest rates and reduces inflationary expectations.

Inflation Controlling Measures

Supply-Side Measures

These long-term strategies aim to increase the economy's productive capacity, allowing it to meet demand without price hikes.

- **Incentivizing Production:** Providing subsidies or tax breaks for essential industries (e.g., renewable energy or semiconductors) to lower production costs.
- **Labor Market Reforms:** Improving education and vocational training to increase worker productivity and reduce labor shortages.
- **Deregulation:** Removing "red tape" to encourage competition and lower business costs.
- **Infrastructure Investment:** Improving transport and energy networks to reduce logistical costs and increase efficiency

Inflation Controlling Measures

Other Targeted Actions

- **Exchange Rate Intervention:** A central bank may target a higher exchange rate to make imports cheaper, directly lowering the cost of foreign goods.
- **Price Controls or Caps:** In extreme cases, governments may impose price ceilings on essential goods like energy or food to protect vulnerable citizens.
- **Anti-Hoarding Laws:** Restrictions on stockpiling essential commodities can prevent artificial supply shortages.

Impact of Inflation on Economic Evaluation

In recent years, inflation remains a critical variable in economic evaluation, influencing how projects, investments, and national growth are assessed. It primarily distorts the time value of money, requiring evaluators to distinguish between nominal and real values to avoid overestimating or underestimating project returns by as much as 50% or more. The evaluation of long-term projects is heavily influenced by how inflation interacts with cash flows and discount rates:

Real vs. Nominal Analysis: Evaluations must use consistent data. If cash flows include expected inflation (nominal), they must be discounted using a nominal interest rate. If they exclude inflation (real), a real discount rate must be used.

Purchasing Power Erosion: Future cash receipts lose value over time. For example, at 5% annual inflation, a \$100 receipt in year three is worth only \$86.40 in today's terms.

Cost Overruns: In sectors like construction, shifting inflation rates cause significant deviations in the price of goods and services, necessitating frequent budget revisions and advanced forecasting models to avoid excessive liability.

Distorted Price Signals: High inflation creates "inflationary noise," making it difficult for businesses to distinguish between shifts in market demand and general price increases, often leading to inefficient resource allocation.

Impact of Inflation on Economic Evaluation

Effects on Economic Growth & Stability

Economic evaluations often look for a "threshold" where inflation transitions from helpful to harmful:

Growth Thresholds: Studies suggest an optimum inflation rate (e.g., around 6% for developing economies like Nepal) supports growth, while exceeding this threshold can jeopardize economic stability and increase poverty.

Investment and Savings: High inflation lowers the real value of savings, discouraging long-term capital accumulation. Conversely, modest, predictable inflation can stimulate activity by encouraging immediate spending and reducing the real burden of existing debt.

International Competitiveness: If a country's inflation significantly exceeds that of its trading partners, its exports become less competitive, potentially worsening the current account balance and slowing growth.

Impact of Inflation on Economic Evaluation

Global Outlook for 2026

Projections for 2026 indicate a global move toward stabilization, though risks remain:

Inflation Targets: Global headline inflation is projected to fall to 3.7%–3.8%.

Monetary Policy: Central banks, such as the RBA and the Fed, are expected to maintain tighter policies through 2026 to ensure inflation returns to long-term targets (typically 2%).

Regional Trends:

United States: Inflation is expected to slow to 2.7% .

Euro Area: HICP inflation is projected to reach approximately 1.7% by early 2026.

Emerging Markets: Growth in low-income countries is projected to rise to 5.7% in 2026 as inflationary pressures ease.

Real and nominal exchange rates, fiscal budget and monetary policy

An **exchange rate** is the price of one country's currency in terms of another country's currency.

If 1 US dollar = NRs. 133, it means to buy 1 dollar, one needs 133 Nepali rupees.

Exchange rates are very important in international trade, imports, exports, tourism, and investment.

The **nominal exchange rate** is the actual market rate at which one currency can be exchanged for another.

It tells us how many units of domestic currency are needed to buy one unit of foreign currency, or vice versa. It only shows the money value of one currency against another. It does not show the difference in purchasing power between countries.

$$\text{Nominal Exchange Rate} = \frac{\text{Domestic Currency}}{\text{Foreign Currency}}$$

Real and nominal exchange rates, fiscal budget and monetary policy

Real Exchange Rate: The real exchange rate measures the relative price of goods and services between two countries after adjusting for price levels.

It tells us how many units of domestic goods must be given up to obtain foreign goods. the real exchange rate shows whether foreign goods are cheap or expensive compared to domestic goods.

$$\text{Real Exchange Rate} = \text{Nominal Exchange Rate} \times \frac{\text{Foreign Price Level}}{\text{Domestic Price Level}}$$

or

$$RER = e \times \frac{P_f}{P_d}$$

Where:

- e = nominal exchange rate
- P_f = foreign price level
- P_d = domestic price level

Real and nominal exchange rates, fiscal budget and monetary policy

- If the real exchange rate rises, foreign goods become relatively more expensive, or domestic goods become relatively cheaper.
- If the real exchange rate falls, foreign goods become relatively cheaper, or domestic goods become relatively more expensive.

Suppose:

- Nominal exchange rate = 133 NRs/USD
- Price of a basket in USA = \$10
- Price of similar basket in Nepal = NRs. 1000

Then:

$$RER = 133 \times \frac{10}{1000}$$

$$RER = \frac{1330}{1000} = 1.33$$

Real and nominal exchange rates, fiscal budget and monetary policy

- This means 1.33 units of Nepali goods are exchanged for 1 unit of American goods. So, US goods are relatively more expensive compared to Nepalese goods.

A **fiscal budget** is the government's annual financial plan showing:

- expected revenue
- expected expenditure
- borrowing and repayment plans
- It is usually prepared for one fiscal year.

It is the income and spending statement of the government.

Importance of Fiscal Budget

- helps allocate resources
- supports economic development
- funds infrastructure

Real and nominal exchange rates, fiscal budget and monetary policy

- reduces unemployment
- maintains stability
- supports social welfare
- encourages balanced growth

Monetary policy is the policy by which the central bank controls the money supply, interest rate, and credit conditions to achieve macroeconomic goals. It controls how much money and credit are available in the economy. In Nepal, it is conducted by Nepal Rastra Bank.

Objectives of Monetary Policy

- price stability
- control of inflation
- economic growth
- employment generation
- exchange rate stability
- financial stability
- balance of payments stability

Real and nominal exchange rates, fiscal budget and monetary policy

Basis	Fiscal Policy	Monetary Policy
Definition	Policy of government related to taxation and public expenditure to influence the economy	Policy of central bank to control money supply, credit, and interest rates
Authority / Control	Government / Ministry of Finance	Central Bank (e.g., Nepal Rastra Bank)
Main Objective	Economic growth, employment, income redistribution, development	Price stability, inflation control, financial stability
Focus Area	Government revenue and spending	Money supply and credit conditions
Key Instruments	Taxation, public expenditure, subsidies, borrowing, budget deficit/surplus	Bank rate, repo rate, CRR, SLR, Open Market Operations (OMO)
Nature of Control	Direct (government directly spends or taxes)	Indirect (influences banks and financial system)
Time Lag	Long (due to political approval and implementation delay)	Short (central bank can act quickly)
Flexibility	Less flexible	More flexible
Impact on Demand	Directly affects aggregate demand ($G + T$)	Indirectly affects demand via interest rate and credit
Impact on Inflation	Controls inflation through taxation and spending	Controls inflation by reducing money supply
Impact on Employment	Can directly create jobs (infrastructure projects)	Indirect effect through investment and growth
Use in Recession	Increase spending, reduce taxes (Expansionary Fiscal Policy)	Lower interest rates, increase money supply

Real and nominal exchange rates, fiscal budget and monetary policy

Use in Inflation	Reduce spending, increase taxes (Contractionary Fiscal Policy)	Increase interest rates, reduce money supply
Budget Involvement	Yes (annual budget is core tool)	No direct budget involvement
Political Influence	High (depends on government decisions)	Low (central bank often independent)
Public Debt Impact	Can increase debt through deficit financing	Does not directly create public debt
Effect on Income Distribution	Can reduce inequality via progressive taxes and subsidies	No direct role in income distribution
Example (Nepal)	Government builds roads, reduces taxes	Nepal Rastra Bank changes bank rate or CRR
Risk / Limitation	Political delays, budget deficits, inefficiency	May not work well if banks don't lend (liquidity trap)
Scope	Broad (affects entire economy and development)	Narrower (focus on financial system and liquidity)
Type of Policy	Expansionary / Contractionary	Expansionary / Contractionary
Effect on Investment	Direct via government projects	Indirect via interest rate changes
Control over Economy	Structural + developmental	Stabilization + liquidity control

Financial Statement

A **financial statement** is a formal record that shows the financial performance and financial position of a business or organization over a specific period of time.

It helps managers, investors, and stakeholders understand:

- how much the company earns
- how much it spends
- what it owns
- what it owes

Financial statements are usually prepared annually or quarterly.

Financial statements are classified into income statement, balance sheet, cash flow statement, statement of owner's equity, and statement of retained earnings. These statements help evaluate profitability, financial position, and cash flow of a business.

Financial Statement

Type of Financial Statement	Definition	Main Components	Purpose
Income Statement (Profit and Loss Statement)	A financial statement that shows the revenue, expenses, and profit or loss of a business during a specific period of time.	Revenue, Cost of goods sold, Operating expenses, Taxes, Net profit	To determine whether the company is making profit or loss during the accounting period.
Balance Sheet	A financial statement that shows the financial position of a business at a specific point in time.	Assets, Liabilities, Owner's equity	To show what the company owns and what it owes.
Cash Flow Statement	A financial statement that records the inflow and outflow of cash in a business during a given period.	Operating activities, Investing activities, Financing activities	To evaluate the liquidity and cash availability of the business.
Statement of Owner's Equity	A statement that shows changes in the owner's capital or shareholders' equity over a period of time.	Beginning capital, Additional investment, Net income, Withdrawals	To show how owner's investment and profits change the company's equity.
Statement of Retained Earnings	A financial statement showing how profits are retained or distributed as dividends in a company.	Beginning retained earnings, Net income, Dividends, Ending retained earnings	To show how profits are used within the company.

Thank you for your attention!